# When Should Robots Hide Where They Wait? Leakage-Aware Randomized Staging for Capacity-Constrained Multi-Robot Response

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***Abstract*—Security or inspection robots wait at staging docks and must answer a burst of alarms placed by an adversary. Randomizing where they wait helps only if the adversary cannot see the realized staging. We model this as a zero-sum game in which the planner commits to a distribution over dock configurations, the adversary observes the realized configuration with probability $q$, and the planner answers with a capacity-limited, all-or-nothing assignment over path-time reachability. We show that per-site marginals do not determine the value, so marginal-coverage formulations do not apply; that the value is convex in $q$ with a threshold $q^\star$ beyond which randomization loses its strict worst-case advantage; and that $q^\star$ is computable by one linear-fractional program. An exact solver with a joint-recourse branch-and-bound separation handles $3.3 \times 10^{11}$ attacks in seconds. In 540 exact games the benefit concentrates where reachability is sparse and capacity tight. In closed-loop simulation, optimized randomization beats the best fixed staging by 8.1%, 5.8% and 2.6% at $q = 0, 0.25, 0.5$, while uniform randomization loses 12–18%. In concurrent three-robot Gazebo/Nav2 execution of every dispatch on 15 layouts, planning on navigation-verified reachability realizes exactly the planned value in 12 layouts, whereas unverified reachability delivers under half of it.**

***Index Terms*—Multi-Robot Systems; Planning, Scheduling and Coordination; Surveillance Robotic Systems; Planning under Uncertainty.**

## I. Introduction

Mobile robots that patrol and respond inside warehouses, plants and campuses wait at staging docks between incidents and are dispatched when alarms fire. Where they wait determines what they can reach in time. If alarms are caused by a strategic adversary—an intruder triggering sensors to overload the fleet, or a saboteur disabling several subsystems at once—the adversary loads the zones the fleet is least able to serve, and a fixed staging plan, once observed, reveals exactly where the gaps are.

Randomizing the staging configuration from shift to shift is the obvious remedy, and randomized defence is the core idea of security games [1], [2]. Two features of robot response are not captured there. First, a zone is restored only if *all* of its alarms are served, and robots at a dock are shared among zones, so the payoff depends on which docks are open *together*, not on how often each is open. Second, staging is often partly visible, so randomization is only as valuable as it is concealed. Our contributions are:

1) A **leakage-aware staging game** with capacity-coupled, all-or-nothing response over path-time reachability (Sec. III).
2) **Structural results**: per-site marginals are insufficient (Lemma 2, Fig. 1); the value is convex in $q$ with a threshold $q^\star$ (Prop. 1) computable by one linear-fractional program, with a certified bound and free randomization among ties (Prop. 2, Cors. 1–2).
3) An **exact solver** whose joint-recourse separation was about 2700 times faster than enumerated separation on our 40-zone instances (Sec. V).
4) **Evidence on when randomization pays**: 540 exact games, closed-loop simulation on synthetic maps and a real building's map, sensitivity analyses, and concurrent three-robot Gazebo/Nav2 execution (Sec. VI).

The practical message, valid for all-or-nothing restoration, is a deployment rule: randomize staging only where reachability is sparse and capacity tight, only if the probability that the adversary observes the realized staging is below $q^\star$, and only on reachability verified with the deployed navigation stack; never randomize uniformly.

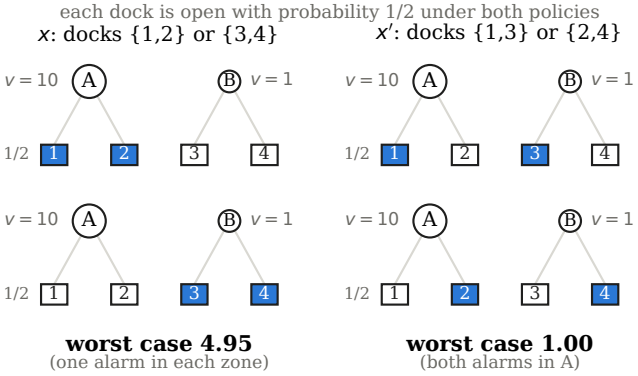



Fig. 1. Same marginals, different worst case (Lemma 2). A ($v = 10$) is reachable from docks 1–2, B ($v = 1$) from 3–4; $m = 2$, $c = 1$, $n = 2$, $p = 0.9$; filled docks are open. Both policies open every dock with probability $\frac{1}{2}$, but only $x$ ever opens both docks that reach A, so only $x$ can absorb two alarms there.

## II. Related Work

Adversarial multi-robot patrolling randomizes *routes* so that an intruder cannot predict coverage [3], [4]; these models ask

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TABLE I
CLOSEST PRIOR MODELS. RAND.: RANDOMIZED FIRST STAGE. LEAK.: REALIZATION OBSERVED WITH PROBABILITY $q \in (0,1)$. COUPLED: RESPONSE COUPLES TARGETS. MARG.: VALUE DETERMINED BY PER-TARGET MARGINALS ($^\dagger$LEMMA 2).

| Model | Rand. | Leak. | Coupled | Marg. | exact $q^\star$ |
|---|---|---|---|---|---|
| Patrolling games [3], [4] | ✓ | – | – | – | – |
| Security games [5], [2] | ✓ | – | – | ✓ | – |
| Leakage security games [8] | ✓ | ✓ | – | ✓ | – |
| DAD, robust siting [11], [9] | – | $q$=1 | ✓ | – | – |
| Randomized robust [12] | ✓ | – | ✓ | – | – |
| This work | ✓ | ✓ | ✓ | –$^\dagger$ | ✓ |

whether a target is visited in time, not whether a capacity-limited fleet can jointly answer a multi-alarm burst. Stackelberg security games compute randomized resource allocations [1], [5], solved at scale by double oracle [6], [7]; Xu et al. [8] let the attacker observe part of the realized deployment. Their efficient formulations rely on per-target marginal coverage, which does not determine our payoff (Lemma 2). Our place–attack–assign structure is adapted from air-defence battery siting [9], [10] and defender–attacker–defender models [11], which assume a fully observed deployment ($q = 1$), exactly where randomization has no strict advantage (Prop. 1); that random first-stage decisions can help in discrete robust problems is known [12]. Table I summarizes: each ingredient exists separately, but no prior model combines a randomized first stage, partial observation and target-coupled recourse. The combination changes the problem class: leakage-aware security games stay tractable *because* payoffs depend only on marginals, and the defender–attacker–defender view loses the benefit of randomization altogether. What it yields here is an exact threshold, a separation oracle that keeps the configuration-space LP exact at facility scale, and evidence, including negative results, on when the benefit survives navigation. MEXCLP [13] serves as a fixed-placement baseline.

## III. PROBLEM FORMULATION

A facility map has $K$ candidate staging sites $I$ and $N$ zones $J$ with priorities $v_j > 0$. Site $i$ *reaches* zone $j$ ($a_{ij} = 1$) if $\tau_{ij} = \ell_{ij}/s + \sigma \le T$, with obstacle-aware path length $\ell_{ij}$, speed $s$, service time $\sigma$ and deadline $T$. The planner opens $m$ docks of $c$ robots each; a configuration is $d \in \mathcal{D} = \{d \subseteq I : |d| = m\}$, and a policy $x \in \Delta(\mathcal{D})$ is re-sampled every shift. The adversary places $n$ alarms, $w \in \mathcal{W} = \{w \in \mathbb{Z}^N_{\ge 0} : \sum_j w_j = n\}$, knowing $x$ and, with probability $q$, the realized $d$. Each robot serves at most one alarm and succeeds independently with probability $p$; zone $j$ is restored only if all $w_j$ alarms succeed, as when an intrusion is contained only once every triggered entry point is covered. The terminal value is

$$U(d,w) = \sum_{j:w_j=0} v_j + \max_{S \in \mathcal{F}(d,w)} \sum_{j \in S} v_j\, p^{w_j}, \qquad (1)$$

where $\mathcal{F}(d,w)$ contains the loaded-zone sets that robots at open docks can serve with exactly $w_j$ robots each, respecting reachability and capacity. With $\mu_d = \min_w U(d,w)$ and best fixed value $P = \max_d \mu_d$,

$$V_x(q) = (1-q) \underbrace{\min_w \sum_d x_d U(d,w)}_{z(x)} + q \underbrace{\sum_d x_d \mu_d}_{b(x)}, \qquad (2)$$

$V^\star(q) = \max_x V_x(q)$, and the *concealment gain* is $G(q) = (V^\star(q) - P)/P$.

## IV. STRUCTURAL RESULTS

**Lemma 1** (Terminal feasibility)**.** *$S \in \mathcal{F}(d,w)$ iff for every $T \subseteq S$, $\sum_{j \in T} w_j \le c\,|\{i \in d : \exists j \in T,\ a_{ij} = 1\}|$.*

*Proof.* Feasibility is a flow from docks (capacity $c$) over uncapacitated reachability arcs to zones (demand $w_j$); every finite cut is a zone set $T$ with its reachable docks, so max-flow/min-cut gives Gale's condition [14]. □

With at most $n$ loaded zones, (1) is evaluated in $O(2^L L)$, $L \le n$, by a subset-AND transform.

**Lemma 2** (Marginals are insufficient)**.** *There exist policies with identical per-site opening probabilities but different values $V_x(0) \ne V_{x'}(0)$.*

*Proof.* Take Fig. 1: $x$ opens $\{1,2\}$ or $\{3,4\}$ and $x'$ opens $\{1,3\}$ or $\{2,4\}$, each with probability $\frac{1}{2}$. Against $x$, the attack $(2,0)$ yields $\frac{1}{2}(1+10p^2) + \frac{1}{2}\cdot 1 = 5.05$, $(1,1)$ yields $\frac{1}{2}(10p + p) = 4.95$, and $(0,2)$ yields $10.405$, so $z(x) = 4.95$. Under $x'$ zone A never receives two robots, so $(2,0)$ yields $1 = z(x')$. □

The decision variable is therefore the joint distribution, which rules out per-site-marginal formulations such as [8] (though not every compact representation). The gap is common: in 116 of the 321 grid games of Sec. VI with a concealment gain, some distribution with the optimal marginals is worse than the best fixed configuration, and realizing the optimal marginals by standard comb sampling falls below fixed staging in 59.

**Proposition 1** (Convexity and threshold)**.** *$V^\star(q)$ is convex, piecewise linear and non-increasing with $V^\star \ge P = V^\star(1)$. If $G(0) > 0$ there is $q^\star \in (0,1]$ with $V^\star(q) > P$ iff $q < q^\star$; if $G(0) = 0$ we set $q^\star = 0$.*

*Proof.* Each $V_x$ is affine in $q$ with slope $b(x) - z(x) \le 0$, and $V^\star$ is their pointwise maximum. Pure configurations give $V^\star \ge P$, and $V_x(1) = b(x) \le P$. □

**Proposition 2** (Exact threshold by one LP)**.** *Let $\mathcal{D}^\star = \arg\max_d \mu_d$. If a mixture within $\mathcal{D}^\star$ has $z(x) > P$, then $q^\star = 1$. Otherwise $q^\star = t^\star/(1+t^\star)$ with*

$$t^\star = \sup_{x \in \Delta(\mathcal{D}),\, b(x) < P} \frac{z(x) - P}{P - b(x)}, \qquad (3)$$

*which the Charnes–Cooper substitution $y = \lambda x$, $\zeta = \lambda z$ [15] turns into the LP $\max\{\zeta - P\lambda : \zeta \le \sum_d y_d U(d,w)\ \forall w;\ P\lambda - \sum_d y_d \mu_d = 1;\ \sum_d y_d = \lambda;\ y, \lambda \ge 0\}$.*

*Proof.* $V_x(q) - P = (1-q)(z(x) - P) - q(P - b(x))$. If $b(x) = P$, $x$ lives on $\mathcal{D}^\star$ and the difference is positive for all

$q < 1$ iff $z(x) > P$; otherwise it is positive iff $q/(1-q) < (z(x)-P)/(P-b(x))$. □

**Corollary 1** (Certified bound)**.** *For the concealed optimum $x^0$ with $z_0 = z(x^0) > P$ and $b_0 = b(x^0)$, $q^\star \geq (z_0 - P)/(z_0 - b_0)$.*

**Corollary 2** (Free randomization among ties)**.** *A mixture of configurations attaining $P$ with $z > P$ weakly dominates every fixed configuration for all $q$, strictly for $q < 1$.*

*Remark* 1 (Scope). $q^\star$ marks where the *strict* worst-case advantage disappears; fixed staging is then optimal but not necessarily unique. Randomized policies are not thereby worse under criteria the objective omits, and mixtures over co-optimal configurations remain optimal for every $q$ (Cor. 2). Leakage is treated as exogenous. If configurations are re-drawn each shift of length $H$ and transit between docks is observable, $q_{\text{eff}}(x) = q + (1-q)\mathbb{E}_x[\tau]/H$ with $\mathbb{E}_x[\tau] = \sum_{d,d'} x_d x_{d'} \tau(d,d')$ is quadratic in $x$. It multiplies the piecewise-linear $z(x) - b(x)$, so the end-to-end objective $z(x) - q_{\text{eff}}(x)(z(x) - b(x))$ is piecewise cubic and not concave in general, and Prop. 2 no longer applies (evaluated in Sec. VI-B). An attacker who times attacks to relocation should be modelled with a larger $q$.

## V. Exact Computation

$V^\star(q)$ and $t^\star$ are LPs with one constraint per attack, $|\mathcal{W}| = \binom{N+n-1}{n}$, which we solve by constraint generation. The separation problem $\min_w \sum_d x_d U(d,w)$ is the hard step: in marginal-coverage games the attacker's best response decomposes across targets, but here zones are coupled through shared docks, so no decomposition exists. We solve it by exact depth-first branch-and-bound over allocations, zones in decreasing value, with the *joint-recourse bound* $\sum_d x_d \big[\sum_{j \in F, w_j = 0} v_j + \max_{S \subseteq F_+, S \in \mathcal{F}} \sum_{j \in S} v_j p^{w_j}\big] + \sum_{j \notin F} v_j - \text{top}_r\{v_j : j \notin F\}$ at a node fixing zones $F$ (loaded subset $F_+$) with $r$ alarms left. It is admissible for every configuration at once, since extra load outside $F$ cannot invalidate a feasible service of $S$ and at most $r$ further zones lose value. For $q > 0$ exact $\mu_d$ is computed lazily, only for rows that receive weight or could be a best response, so termination certifies the exact value. Prop. 2 replaces bisection over $q$, which needs one full solve per step and only brackets $q^\star$.

**Validation and ablation.** The evaluator matched max-flow exactly and an MILP to $3.6 \times 10^{-15}$ on 810 random pairs; re-optimized LPs at $q^\star \pm 0.01$ confirmed every threshold on 26 instances; the matrix-free solver reproduced $P$ and $V^\star(q)$ on 24 grid instances to $3.6 \times 10^{-15}$, and its best response matched enumeration of 177,100 allocations on 36 random mixtures. Table II compares three exact methods on identical instances; they agree to $4 \times 10^{-15}$. Branch-and-bound time stays near 0.1 s with about ten best-response calls, about 2700 times faster than enumerated separation at 40 zones. This is empirical for these instances (sparse, skewed priorities), not a complexity guarantee; uniform priorities prune less. The linear-fractional LP took 8–56 ms against 0.27–0.40 s for a 10-step bisection, and unlike bisection certifies the tie case $q^\star = 1$. Computations are single-threaded Python/numba on one 2.1 GHz core; intervals are 95% $t$-intervals over maps or layouts.

TABLE II
Solver ablation: median time (s) over three maps for $P$ and $V^\star(0)$ ($K = 12$, $m = 3$, $n = 6$, $c = 1$); "mem": matrix over 1.2 GB; limit 600 s.

| $N$ | $\lvert\mathcal{W}\rvert$ | full matrix | enumerated CG | B&B CG (ours) |
|---|---|---|---|---|
| 12 | 1.2e4 | 2.27 | 0.38 | 0.08 |
| 20 | 1.8e5 | 42.0 | 4.66 | 0.15 |
| 25 | 5.9e5 | 158 | 17.8 | 0.11 |
| 30 | 1.6e6 | mem | 46.7 | 0.11 |
| 40 | 8.1e6 | mem | 297 | 0.11 |
| 50 | 2.9e7 | mem | $> 600$ | 0.14 |

## VI. Experiments

### A. Exact games and scaling

The grid study is exploratory, with design and continue/stop rule fixed in a dated protocol file before any $m \geq 3$ result [16]; the simulation's sparse regime was chosen from its passing cell, and the remaining studies were added afterwards and are reported in full. The grid crosses $m \in \{2,3,4\}$, $c \in \{1,2,3\}$, reachability density $\{0.20, 0.35, 0.60\}$ and uniform or Zipf-skewed priorities ($K = N = 12$, $n = 6$, $p = 0.7$, 10 maps per cell): 540 exact games (Fig. **??**). Three cells passed the rule ($G(0) \geq 5\%$, CI excluding zero, $q^\star \geq 0.25$): $(m,c) = (3,1)$ sparse skewed ($12.0 \pm 5.4\%$, $q^\star = 0.78$), $(3,2)$ sparse uniform ($5.4 \pm 2.6\%$) and $(4,1)$ sparse skewed ($7.1 \pm 3.6\%$). Dense reachability never exceeded $1.2\%$, since one configuration then covers every attack, and mean $G(0)$ falls from $3.8\%$ to $1.8\%$ as $m$ goes from 2 to 4. The re-optimized $G(q)$ curves are convex, as Prop. 1 predicts, and most of the benefit is lost by $q \approx 0.5$ (Fig. **??**). Of 321 instances with a gain, 97 had $q^\star = 1$ through ties (Cor. 2); the bound of Cor. 1 held in all 540 and recovered 68% of $q^\star$ on average. Optimal supports were small (mean 4.0 configurations). On 54 larger instances ($K = 20$, $N = 30$–$50$, up to $2.9 \times 10^7$ attacks, never enumerated) every $c = 1$ instance kept a gain ($G(0) = 7.8\%$ at $N = 30$, $5.2\%$ at $N = 50$), solved in a median 0.1–0.3 s; a stress case with $3.3 \times 10^{11}$ attacks solved in 11.5 s.

### B. Closed-loop simulation and sensitivity

Three $120 \times 80$ m maps (warehouse, office, campus; Fig. **??**), five layouts each with 12 sites and 12 zones, robots at 1.5 m/s with lognormal speed noise ($\sigma = 0.15$) and 30 s service, and deadlines giving reachability density 0.20 (sparse) or 0.60 (dense control). Each 8 h shift draws a configuration, robots relocate along shortest paths, and an attack at a uniform time is observed if robots are in transit or with probability $q$; robots then drive from their actual positions. Five policies (best fixed, MEXCLP, uniform, concealed $x^0$, leakage-aware for the true $q$) share paired random draws over 1000 shifts per layout. In the sparse regime (Fig. **??**) the leakage-aware policy beats fixed staging by $8.1 \pm 2.7\%$ (15/15 layouts), $5.8 \pm 2.3\%$ (13/15) and $2.6 \pm 2.1\%$ (10/15) at $q = 0, 0.25, 0.5$; at $q = 0.5$ the unadapted $x^0$ keeps only $0.9\%$. Re-optimizing for leakage is therefore what preserves the benefit. Uniform randomization loses 12–18%, since it spends probability on configurations with large gaps; the MEXCLP placement is within 1% of fixed maximin. In the dense control regime the optimized policies

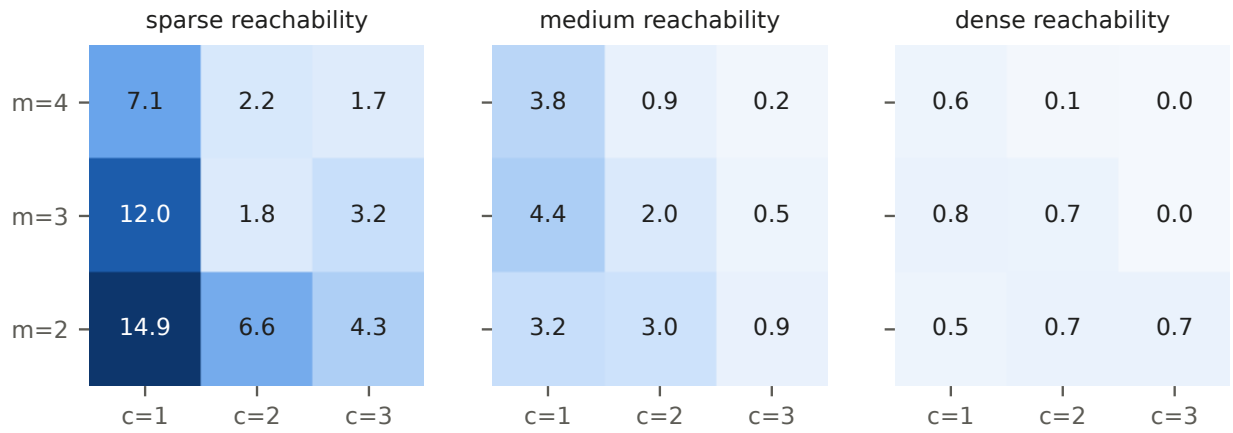



Fig. 2. Mean concealment gain $G(0)$ (%) over 10 maps per cell, skewed priorities. Randomization pays only with sparse reachability and tight capacity ($c = 1$), and the gain shrinks with more docks.

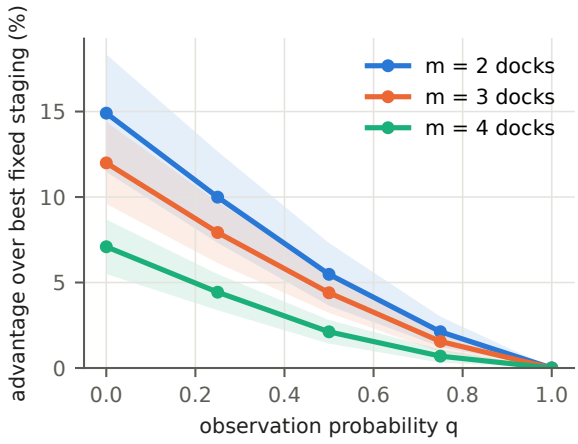



Fig. 3. Re-optimized advantage over the best fixed staging versus observation probability $q$ (sparse, skewed, $c = 1$; mean ± s.e., 10 maps). Each instance's curve is convex and reaches zero at its own $q^\star$; the mean reaches zero only at $q = 1$ because $q^\star$ varies across maps.

are indistinguishable from fixed staging, as Fig. **??** predicts, while uniform randomization still loses 3.5–5.6%. Simulated values track the model ($r = 0.955$) and are on average 3.1% lower, because speed noise and mid-relocation positions cause deadline misses the model omits.

On the map of a real building in simulation (no physical robots), the Willow Garage office (0.1 m laser map, obstacles inflated by the robot radius), exact gains are 10.7, 7.0, 4.2% (synthetic 11.2, 7.5, 4.2%), with mean $q^\star = 0.82$ (synthetic 0.83), and closed-loop gains 7.3, 6.6, 5.5% (4/5 layouts; intervals include zero with five layouts); uniform randomization loses 14–20% in every layout. Relocation exposure (Remark 1) is negligible for shifts of 15 min or longer (gain 11.2 → 10.6%; in 15-minute closed-loop shifts at $q = 0.25$, +5.6%) but matters at 2 min (6.4%), where exposure-aware re-optimization recovers at most 1.1 points. The benefit is specific to all-or-nothing restoration with reliable robots: in the favourable cell $G(0)$ is 9.5% at $p = 0.7$, 15.4% at $p = 0.9$, 2.8% at $p = 0.5$, and 2.1% under partial credit, where a zone earns $v_j$ times its fraction of resolved alarms; per-instance gains under the two rules correlate only at $r = 0.48$. It also falls to 3–4% with uniform priorities, density 0.45 or $n = 4$.

TABLE III
CONCURRENT THREE-ROBOT NAV2 EXECUTION (15 LAYOUTS, $q = 0.25$). GAINS VS. BEST FIXED STAGING (%, MEAN ± 95% CI); "EXACT": LAYOUTS WHERE NAV2 REALIZES EXACTLY THE PLANNED VALUE. THE SLACK ROUND RAN THREE TIMES; ‡EXCLUDES THE LAYOUT WHOSE FIXED BASELINE ITSELF FAILS.

| reachability | disp. (conc.) | on time | planned | Nav2 | gain>0 | exact |
|---|---|---|---|---|---|---|
| model | 79 (58) | 128/157 | $7.5 \pm 2.5$ | $3.3 \pm 4.9$ | 10/15 | 4/15 |
| + slack $\geq 3.8$ s | 156 (81) | 245/261 | $5.3 \pm 2.5$ | $5.7 \pm 3.9$ | 13/15 | 12/15 |
| 14 layouts‡ | | | $5.4 \pm 2.7$ | $\mathbf{4.1 \pm 2.1}$ | 12/14 | 12/14 |
| + shared-zone margin | 22 (10) | 32/35 | $3.3 \pm 2.6$ | $2.0 \pm 1.4$ | 8/15 | 13/15 |

### C. Gazebo/Nav2 with concurrent robots

Each map was built in Gazebo Harmonic with Jackal-scale ($0.5 \times 0.4$ m) robots running ROS 2 Nav2 (NavFn, Regulated Pure Pursuit at 1.5 m/s). On 92 single-robot response legs, Nav2 time $= 0.997\, t_{\text{model}} + 1.0$ s ($R^2 = 0.995$; 95th-percentile error 3.8 s), the model and Nav2 classify 91 of 92 pairs identically as reachable, and reachability rebuilt from Nav2-calibrated times gives gains of 10.7, 7.2, 4.0% (model 11.2, 7.5, 4.2%). We then executed in Gazebo every dispatch that the leakage-aware policy ($q = 0.25$) or best fixed staging can produce on all 15 layouts, with three Nav2 stacks running throughout (real-time factor 0.5). Robots perceive each other through a virtual scan built from shared poses in every costmap, and robots sent to one zone use service points 1.4 m apart. A policy's value is the exact expectation over its distribution and success draws, with Nav2 deciding which legs are on time, so there is no sampling noise.

Optimizing on an inaccurate reachability graph can erase the benefit, and correcting the graph recovers most of it (Table III, Fig. 2). With model reachability, Nav2 delivered under half the planned gain, and one layout lost 19%; 26 of 29 failed legs lay on arcs with under 3.8 s of slack. Dropping such arcs before

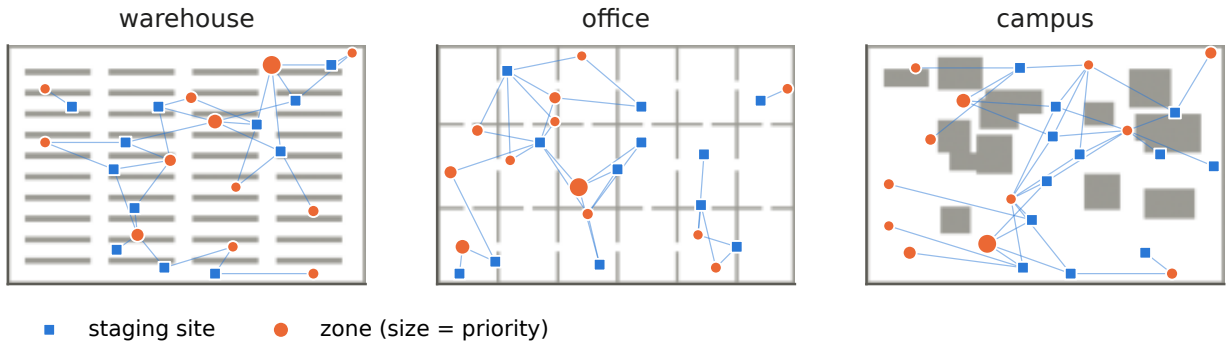



Fig. 4. Simulation maps (layout 0). Lines mark site–zone pairs reachable within the deadline along obstacle-aware shortest paths (sparse regime).

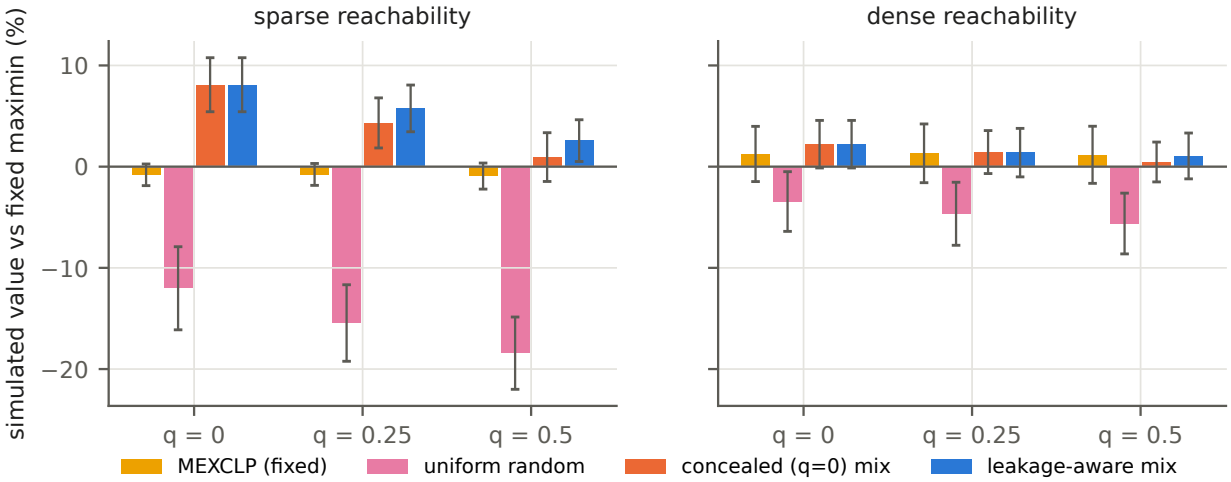



Fig. 5. Closed-loop simulation: mean change in restored value relative to the best fixed staging (95% CI over 15 layouts; 1000 shifts each).

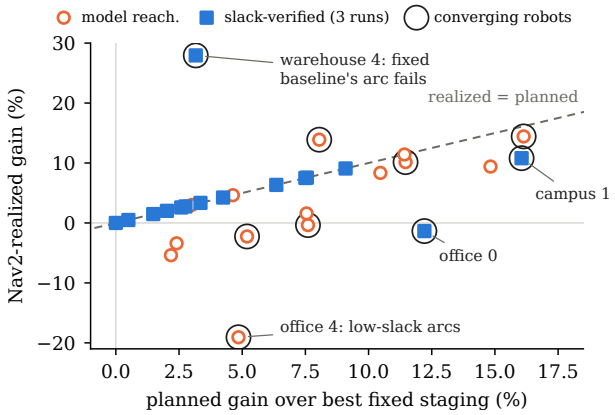



Fig. 6. Planned vs. Nav2-realized gain per layout, model reachability (circles) and slack-verified (squares, mean of three runs); rings mark converging-robot failures. Slack-verified: $4.1 \pm 2.1\%$ vs. $5.4 \pm 2.7\%$ planned on 14 layouts; $5.7 \pm 3.9\%$ on all 15 (raised by warehouse 4).

planning, a rule fixed in advance, makes 12 of 15 layouts realize exactly the planned value in all three repetitions; on the 14 layouts whose fixed baseline succeeds, Nav2 realizes $4.1 \pm 2.1\%$ against $5.4 \pm 2.7\%$ planned. One warehouse layout, whose *fixed* baseline's own arc fails, lifts the 15-layout mean to $5.7\%$; we do not rely on that figure. Timing was nearly deterministic: the median arrival spread across repetitions was 0.09 s, 3 of 87 legs changed on-time status, and only one layout's gain changed. All remaining failures come from two unmodelled mechanisms: robots converging on one zone delay each other (95th percentile 9.6 s vs. 2.7 s for distinct zones, up to 65 s; closest approach 0.49 m between 0.5 m bodies), and one wall-adjacent office arc stalled the controller in all 8 uses. A 9.7 s margin for shared zones removes most interference but leaves little gain (2.0%), so we recommend the slack rule.

## VII. Discussion and Conclusion

**Deployment.** (i) Build reachability with the deployed navigation stack and drop arcs whose slack is below its timing error; (ii) compute the best fixed staging, the optimal mixture and $q^\star$, which varies widely between sites; (iii) deploy the mixture only if a conservative estimate of $q$ is clearly below $q^\star$. If trials support only $q \in [q_\ell, q_h]$, the max-min policy is $x^\star(q_h)$, because $V_x$ is non-increasing in $q$, and minimax regret is one LP with two endpoint constraints, because regret is convex. On the 15 layouts with $q \in [0.2, 0.5]$, planning for $q_h$ guarantees $4.2 \pm 1.8\%$ over fixed staging with at

most $0.8\%$ regret, (minimax regret: $0.5\%$), while planning for $q = 0$ guarantees only $2.8\%$ and falls below fixed staging in 5 layouts.

**Limits.** Conclusions are restricted to all-or-nothing restoration with independent, reliable robots, a rational adversary that knows the policy, and exogenous leakage. Alarm budgets are small ($n \leq 8$), and branch-and-bound effort grows with uniform priorities. Converging robots are the main unmodelled effect; staggered approach routes or interference-aware response are natural extensions. All robot experiments are simulated, with up to three robots. Only one map comes from a real building, and on its cluttered doorways our Nav2 configuration was unreliable (a pilot dispatch aborted or took three times the model time), so it is evaluated in simulation only. The alarm data are synthetic.

Randomized deployment is not inherently safer than fixed placement. Its value depends jointly on which configurations open together, on how much the adversary observes, and on what robots can physically reach under concurrent execution; optimizing one of these without the others produced policies whose advantage vanished. Code, instances, the protocol and all raw results are released at [16].

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